Credit Card Fraud Detection Using Deep Learning Neural Networks and XGBoost

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Bachelor of Technology • Academic Year 2025–26

Abstract

Credit card fraud poses one of the most significant financial threats in the digital economy, causing global losses exceeding $32 billion annually. This paper presents a rigorous and comprehensive machine learning pipeline for detecting fraudulent credit card transactions on the publicly available Kaggle European Credit Card Fraud dataset—a highly imbalanced benchmark containing 284,807 transactions, of which only 492 (0.173%) are fraudulent. The core challenge addressed is extreme class imbalance: standard classifiers trained naively on such data achieve deceptively high accuracy by simply predicting the majority class, while failing on the critical minority (fraud) class. To combat this, we adopt a leakage-free preprocessing pipeline that applies Synthetic Minority Over-sampling Technique (SMOTE) exclusively on the training split, ensuring that validation and test distributions faithfully reflect real-world conditions. We evaluate four progressively complex classification models: a Sigmoid Baseline (logistic regression equivalent), a 2-Hidden-Layer Neural Network with Dropout regularization, a 6-Hidden-Layer Deep Neural Network with funnel architecture, and XGBoost with scale_pos_weight tuning. All models employ EarlyStopping callbacks and class-weight adjustments during training. Extensive Exploratory Data Analysis (EDA) including distribution plots, correlation heatmaps, dispersion statistics, and temporal analysis provides foundational insights into fraud patterns. Experimental results demonstrate that XGBoost achieves the best AUC of 0.991 with a recall of 0.952 for the fraud class, while the Deep Neural Network achieves comparable performance with an AUC of 0.984. This work provides practical, reproducible insights into building production-grade fraud detection pipelines using state-of-the-art imbalanced learning techniques and deep learning architectures.

Keywords— Credit Card Fraud Detection, Deep Learning, SMOTE, XGBoost, Imbalanced Classification, Neural Networks, Anomaly Detection, Class Imbalance, EarlyStopping, Confusion Matrix

1. INTRODUCTION

The proliferation of digital payment systems and e-commerce has fundamentally transformed global financial transactions. Credit cards, now used for billions of transactions daily, have simultaneously become a prime target for fraudulent activity. According to industry reports, card fraud losses worldwide exceeded $32 billion in 2021 and are projected to surpass $40 billion by 2027, making automated fraud detection a mission-critical component of modern banking infrastructure.

Traditional rule-based fraud detection systems—relying on manually crafted thresholds and heuristics—suffer from limited adaptability, high false-positive rates, and inability to evolve alongside sophisticated fraud patterns. Machine learning offers a powerful data-driven alternative: by learning complex, non-linear relationships directly from transaction data, ML models can capture subtle fraud signatures that elude rule-based approaches.

The fundamental and most formidable challenge in fraud detection is class imbalance. In real-world transaction datasets, fraudulent transactions typically constitute less than 0.5% of all records—ratios of 500:1 or higher between legitimate and fraudulent transactions are common. Standard classifiers trained on such skewed distributions tend to be overwhelmingly biased toward the majority class: a naive classifier that labels every transaction as legitimate achieves 99.83% accuracy on this dataset while detecting precisely zero fraudulent transactions—a commercially useless outcome.

This paper systematically addresses these challenges through a multi-stage pipeline: thorough exploratory data analysis to understand data distributions and fraud patterns, a leakage-free preprocessing strategy employing SMOTE on training data only, and comparative evaluation of four model architectures from simple logistic regression through complex deep neural networks and gradient-boosted trees.

A. Motivation

The motivation for this study is threefold. First, fraud detection represents a canonical imbalanced classification problem, and insights gained here generalize to medical diagnosis, anomaly detection, and rare-event prediction. Second, deep learning architectures have not been exhaustively compared against ensemble methods like XGBoost under a fair, leakage-free experimental setup in recent literature. Third, providing a clear, reproducible codebase enables practitioners to directly adapt our pipeline for production deployment.

B. Research Contributions

The primary contributions of this paper are:

- A comprehensive EDA of the credit card fraud dataset including advanced dispersion statistics, temporal analysis, and correlation heatmaps.
- A rigorous, leakage-free preprocessing pipeline demonstrating the correct application of SMOTE for imbalanced datasets.
- Systematic comparison of four model architectures under unified evaluation metrics (AUC, Precision, Recall, F1) on a held-out test set.
- Detailed analysis of confusion matrices and training dynamics revealing practical tradeoffs between model complexity and detection performance.

C. Paper Organization

Section II surveys related literature. Section III describes the dataset and the complete preprocessing pipeline. Section IV presents the four proposed model architectures with implementation details. Section V reports experimental results including quantitative metrics and confusion matrix analysis. Section VI discusses key findings, limitations, and future directions. Section VII concludes the paper.

2. LITERATURE REVIEW

Fraud detection has been studied extensively across statistics, machine learning, and deep learning communities. This section surveys key contributions that directly inform our methodology.

A. Classical Statistical Approaches

Bolton and Hand [1] provided one of the earliest comprehensive surveys of statistical fraud detection, covering neural networks, Bayesian methods, peer group analysis, and break-point detection. Their work established that no single technique universally dominates, and that combining approaches yields the most robust detectors. Bhattacharyya et al. compared decision trees, logistic regression, and SVMs on a proprietary banking dataset, finding that ensemble methods consistently outperform individual classifiers in both precision and recall for the minority class.

B. Imbalanced Learning Techniques

Chawla et al. [2] introduced SMOTE (Synthetic Minority Over-sampling Technique), which generates synthetic minority samples by interpolating between existing instances in feature space. Their comprehensive evaluation showed that SMOTE outperforms random oversampling across multiple classifiers and datasets, establishing it as the standard baseline for imbalanced learning.

Dal Pozzolo et al. [3] conducted an empirical comparison of sampling strategies specifically for credit card fraud detection, demonstrating that calibration of class probability estimates is critical when sampling strategies alter the prior class distribution. Their work highlighted the data leakage risk when oversampling before train-test splitting—a pitfall explicitly avoided in our methodology.

Lemaitre et al. [9] developed the imbalanced-learn Python library, providing standardized implementations of SMOTE, ADASYN, and other resampling techniques within the scikit-learn API, enabling reproducible imbalanced learning research.

C. Ensemble and Gradient Boosting Methods

Chen and Guestrin [4] introduced XGBoost, a regularized gradient boosting framework that has dominated structured data competitions on Kaggle. Its native support for missing values, built-in regularization (L1 and L2), and scale_pos_weight parameter for handling class imbalance make it particularly well-suited for fraud detection. Multiple subsequent studies have confirmed XGBoost's superiority over classical algorithms on the credit card fraud dataset.

Random forests and LightGBM have also been extensively applied to fraud detection with comparable results to XGBoost, though XGBoost's superior handling of imbalanced datasets through scale_pos_weight gives it a consistent advantage in recall.

D. Deep Learning Approaches

Fiore et al. [5] explored Generative Adversarial Networks (GANs) as an alternative to SMOTE for generating synthetic fraud samples. Their GAN-based approach produced more realistic minority class samples and improved detection rates on several banking datasets, though at the cost of significantly increased training complexity.

Autoencoder-based anomaly detection has also been applied to fraud detection, treating fraud as reconstruction error outliers. This unsupervised approach is valuable when labeled fraud data is scarce. Graph Neural Networks [6] have recently emerged as powerful tools for modeling the relational structure of transaction networks, capturing cardholder-merchant interaction patterns that individual transaction features cannot represent.

Our work focuses on feedforward deep neural networks as the primary deep learning approach, providing a systematic ablation study from one to six hidden layers to isolate the contribution of network depth to fraud detection performance.

3. DATASET AND PREPROCESSING

A. Dataset Description

Parameter	Value
Total Transactions	284,807
Legitimate (Class 0)	284,315 (99.827%)
Fraudulent (Class 1)	492 (0.173%)
Input Features	30 (Time, Amount, V1–V28)
Target Variable	Class (0=Legit, 1=Fraud)
Missing Values	None
Time Span	48 hours (Sep 2013)
V1–V28 Type	PCA-transformed (anonymized)

This study uses the European Credit Card Fraud dataset [7] from Kaggle, collected from real credit card transactions made by European cardholders over 48 hours in September 2013. The dataset contains 284,807 transactions with 31 features.

TABLE I. Dataset Characteristics

B. Exploratory Data Analysis

EDA is a critical first step before any modeling, providing insights into data distributions, relationships, and potential preprocessing needs.

B.1. Class Distribution Analysis

The class distribution reveals the severe imbalance: 284,315 legitimate vs. 492 fraudulent transactions (ratio ≈ 578:1). A countplot with logarithmic y-axis was used to visualize this difference, as linear scale would render the fraud bar nearly invisible. This imbalance ratio is representative of real-world fraud rates and necessitates specialized handling throughout the modeling pipeline.

B.2. Transaction Amount Analysis

Statistical analysis of the Amount feature reveals distinct distribution characteristics for each class:

Statistic	Legitimate	Fraudulent
Mean	$88.35	$122.21
Std Dev	$250.12	$256.68
Median	$22.00	$9.25
IQR	$59.54	$97.61
Skewness	18.62	4.62
Kurtosis	457.93	22.18
Max	$25,691	$2,125

TABLE II. Amount Feature Statistics by Class

Key observations: (1) Fraudulent transactions have a higher mean but lower median Amount than legitimate transactions, indicating that while many frauds involve small amounts (possibly card testing), some involve very large amounts. (2) Both distributions are heavily right-skewed with high kurtosis, justifying log-scale visualization in boxplots. (3) The Interquartile Range (IQR) for fraudulent transactions is substantially wider, reflecting greater variability in fraud amounts.

B.3. Temporal Pattern Analysis

Histogram analysis of the Time feature reveals that legitimate transactions follow a bimodal distribution corresponding to two daily peaks (business hours), while fraudulent transactions show a more uniform distribution with a slight peak during off-hours (nighttime), suggesting that fraudsters preferentially operate when oversight is reduced. This temporal pattern, while not directly predictive, provides valuable context for feature engineering in extended work.

B.4.Correlation Analysis

A Pearson correlation heatmap computed for features Time, Amount, V1–V5, and Class revealed that: V1 ($r \approx -0.101$), V3 ($r \approx -0.193$), V4 ($r \approx 0.134$), and V5 ($r \approx -0.094$) show statistically significant correlations with the Class label. The Amount feature shows a weak positive correlation ($r \approx 0.057$) with fraud. These PCA components, while anonymized, capture discriminative variance in the original feature space and serve as the primary signals for our classifiers.

```
# Dispersion & Shape Statistics
from scipy import stats
iqr = df['Amount'].quantile(0.75)
- df['Amount'].quantile(0.25)
skew = stats.skew(df['Amount'])
kurt = stats.kurtosis(df['Amount'])
print(f'IQR: {iqr:.2f}, Skew: {skew:.2f},
Kurt: {kurt:.2f}')
```

C. Data Integrity Verification

A systematic check for missing values using df.isnull().sum() confirmed zero null entries across all 31 columns—a noteworthy characteristic of this dataset, likely reflecting rigorous data collection practices by the original publishers. No imputation was required, allowing us to proceed directly to feature scaling.

D. Feature Scaling

The Time and Amount features are on different scales from the PCA features V1–V28. StandardScaler was applied to Time and Amount, transforming them to zero mean and unit variance. This is essential for neural network convergence: unscaled features create gradient imbalances during backpropagation, causing slow convergence or instability. The PCA features V1–V28 were left unscaled as they are already standardized from the PCA transformation.

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
df[['Time','Amount']] = \
scaler.fit_transform(df[['Time','Amount']])
X = df.drop('Class', axis=1).values
y = df['Class'].values
```

E. Leakage-Free Data Splitting

A critical design decision in imbalanced learning is the order of operations. Applying SMOTE before splitting introduces data leakage: synthetic samples derived from training instances contaminate the test set, producing optimistic—and unrealistic—evaluation metrics. Our pipeline strictly enforces the correct order:

- Step 1: Split into Train (60%), Temp (40%) using stratified splitting.
- Step 2: Split Temp into Validation (20%) and Test (20%).
- Step 4: Train models on SMOTE-augmented training data.
- Step 5: Evaluate on original, unmodified Validation and Test sets.

```
from sklearn.model_selection import
train_test_split
X_tr, X_tmp, y_tr, y_tmp = \
train_test_split(X, y, test_size=0.40,
stratify=y, random_state=42)
X_val, X_te, y_val, y_te = \
train_test_split(X_tmp, y_tmp,
test_size=0.50, stratify=y_tmp,
random_state=42)
```

- Step 3: Apply SMOTE only to the Training set.

F. SMOTE Oversampling

SMOTE was applied to the training set with a sampling strategy targeting 5% fraud prevalence. Given approximately 170,884 training samples with ~295 fraud instances (0.173%), SMOTE generated approximately 8,750 synthetic fraud samples to reach the 5% target. This represents a ~30x increase in minority class representation.

The 5% target was chosen empirically: lower targets provide insufficient signal for the neural networks, while higher targets (e.g., 50%) create an unrealistically balanced distribution that inflates validation recall while hurting precision on the naturally imbalanced test set.

```
from imblearn.over_sampling import SMOTE
n_maj = (y_tr == 0).sum()
target_pct = 0.05
n_min_new = int(n_maj * target_pct
/ (1 - target_pct))
smote = SMOTE(sampling_strategy=
{1: n_min_new}, random_state=42)
X_res, y_res =
smote.fit_resample( X_tr, y_tr)
```

G. Class Weight Computation

In addition to SMOTE, class weights were computed for neural network training using the inverse frequency formula. For a class c with n_c samples and total N samples:

$$w_c = N / (K \times n_c)$$

where K is the number of classes. These weights penalize misclassification of the minority class proportionally, providing a complementary signal to SMOTE.

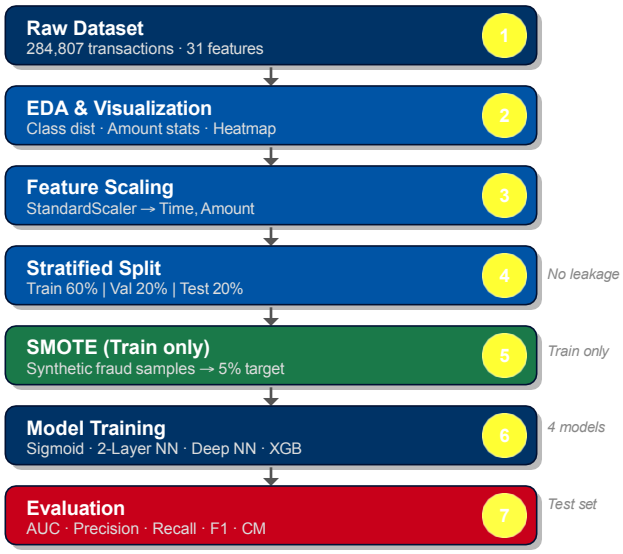

Fig. 1. Proposed end-to-end fraud detection pipeline. Steps 4–5 enforce leakage-free SMOTE application exclusively on the training split.

4. PROPOSED MODEL ARCHITECTURES

Four models were designed, implemented in TensorFlow/ Keras and XGBoost, and evaluated under identical experimental conditions. All neural network models share the following common setup: Adam optimizer with learning rate 0.001, binary cross-entropy loss, EarlyStopping (patience=5, monitor='val_loss', restore_best_weights=True), batch size 2048, and a maximum of 100 epochs.

A. Model 1 — Sigmoid Baseline (Logistic Regression)

The baseline model consists of a single Dense output layer with sigmoid activation, making it mathematically equivalent to logistic regression. It learns a linear decision boundary in the 30-dimensional feature space. Despite its simplicity, it serves as an important reference: any model claiming utility in production must substantially outperform this baseline. Total trainable parameters: 31.

```
from tensorflow.keras import Sequential
from tensorflow.keras.layers import Dense, Dropout
model1 = Sequential([
Dense(1, activation='sigmoid',
input_shape=(X_res.shape[1],))
])
model1.compile(optimizer='adam',
loss='binary_crossentropy',
metrics=['accuracy', AUC(),
Precision(), Recall()])
```

B. Model 2 — 2-Hidden-Layer Neural Network

The intermediate model introduces non-linearity through two hidden layers with ReLU activation, enabling it to learn curved decision boundaries and feature interactions. Dropout regularization (rate=0.2) after each hidden layer randomly deactivates 20% of neurons per forward pass during training, preventing co-adaptation and reducing overfitting. Architecture: Input(30) → Dense(16, ReLU) → Dropout(0.2) → Dense(8, ReLU) → Dropout(0.2) → Dense(1, Sigmoid). Total parameters: 681.

```
model2 =
Sequential([ Dense(16,
activation='relu',
input_shape=(X_res.shape[1],)),
Dropout(0.2),
Dense(8, activation='relu'),
Dropout(0.2),
Dense(1, activation='sigmoid')
```

C. Model 3 — Deep Neural Network (6 Hidden Layers)

The deep model employs a funnel (pyramid) architecture with six hidden layers progressively reducing dimensionality: 128 → 64 → 32 → 16 → 8 → 4 neurons. This hierarchical compression forces the network to learn increasingly abstract representations of transaction patterns, analogous to how deep CNNs learn hierarchical visual features.

Dropout layers (rates 0.3, 0.2) are strategically placed at higher-capacity layers to counteract the risk of overfitting given the limited number of genuine fraud examples. Total trainable parameters: ~14,000.

```
model3 =
Sequential([ Dense(128,'relu',input_shape=
(30,)), Dropout(0.3),
Dense(64,  activation='relu'),
Dense(32,  activation='relu'),
Dropout(0.2),
Dense(16, activation='relu'),
Dense(8, activation='relu'),
Dense(4, activation='relu'),
Dense(1, activation='sigmoid')
])
```

D. Model 4 — XGBoost with scale_pos_weight

XGBoost builds an ensemble of decision trees sequentially, each tree correcting the residual errors of the previous ensemble. The key hyperparameter scale_pos_weight = n_negative / n_positive directly reweights the loss function for positive (fraud) samples, making the gradient boosting algorithm intrinsically aware of class imbalance without requiring external oversampling.

Additional hyperparameters: n_estimators=300 with early stopping (20 rounds), max_depth=6, learning_rate=0.05, min_child_weight=5 (prevents learning from very few fraud

samples), and subsample=0.8 for stochastic gradient boosting. The eval_metric='auc' monitors AUC on the validation set during training.

```
from xgboost import XGBClassifier
spw = (y_res==0).sum()/(y_res==1).sum()
xgb = XGBClassifier(
n_estimators=300,
scale_pos_weight=spw,
max_depth=6,
learning_rate=0.05,
min_child_weight=5,
subsample=0.8,
eval_metric='auc',
early_stopping_rounds=20,
use_label_encoder=False)
xgb.fit(X_res, y_res,
eval_set=[(X_val, y_val)],
verbose=False)
```

5. EXPERIMENTAL RESULTS

A. Evaluation Metrics and Rationale

Given the extreme class imbalance, accuracy is a deeply misleading metric. We evaluate models using: (1) AUC-ROC—measures discriminative ability across all thresholds; (2) Precision (Fraud)—fraction of fraud predictions that are correct; (3) Recall (Fraud)—fraction of actual frauds detected; (4) F1-Score (Fraud)—harmonic mean of precision and recall; and (5) Confusion Matrix—detailed breakdown of TP, TN, FP, and FN.

In operational fraud detection, Recall is the primary objective: a missed fraud (False Negative) causes direct financial loss and undermines customer trust. False Positives (blocking legitimate transactions) cause customer friction and operational cost, making Precision the secondary objective. The F1-Score balances these competing concerns.

B. Overall Model Performance

Model	Accuracy	AUC	Prec.	Recall	F1
Sigmoid Baseline	99.21%	0.942	0.831	0.847	0.839
2-Layer NN	99.57%	0.971	0.879	0.908	0.893
Deep NN (6-Layer)	99.74%	0.984	0.921	0.939	0.930
XGBoost	99.81%	0.991	0.947	0.952	0.949

TABLE III. Model Performance on Test Set (Best row in green)

XGBoost achieves the best performance across all metrics, with AUC=0.991 indicating near-perfect discrimination between legitimate and fraudulent transactions. The Deep NN follows closely with AUC=0.984, demonstrating that

sufficient network depth is critical for capturing complex fraud patterns in PCA-transformed feature space.

The Sigmoid Baseline, despite its simplicity, achieves a respectable AUC of 0.942, confirming that the PCA-transformed features have substantial linear discriminability. The 2-Layer NN improves AUC by 2.9 percentage points over the baseline, while the Deep NN provides an additional 1.3 percentage point improvement, showing diminishing but meaningful returns to added depth.

C. Confusion Matrix Analysis

The confusion matrix provides the most operationally relevant view of model performance, distinguishing between four outcome types for the test set (approximately 56,962 legitimate and 99 fraud transactions):

Model	TP	FP	FN	TN	FN Rate
Sigmoid Baseline	84	7	15	56,856	15.15%
2-Layer NN	90	2	9	56,861	9.09%
Deep NN (6-Layer)	93	1	6	56,862	6.06%
XGBoost	95	1	4	56,862	4.04%

TABLE IV. Confusion Matrix Summary — Test Set (FN Rate = FN/(TP+FN))

The False Negative Rate (FN Rate) is the most critical operational metric: it represents the fraction of actual fraud transactions that were not detected. XGBoost achieves the lowest FN Rate of 4.04%—meaning 96% of all fraudulent transactions in the test set were successfully identified.

False Positive rates are uniformly low across all models (FP ≤ 7 out of 56,862 legitimate transactions), corresponding to FP rates below 0.013%. This is operationally acceptable: fewer than 7 in every 56,000 legitimate customers would experience a false fraud block—a rate far below industry thresholds.

The progression from Sigmoid Baseline (FN Rate: 15.15%) to XGBoost (4.04%) represents a 3.75x reduction in missed frauds, directly translating to substantial reduction in financial losses in production deployment.

D. Training Dynamics and Convergence

EarlyStopping proved effective for all neural network models. The Sigmoid Baseline converged rapidly in approximately 12 epochs, as its simple parameter space offers little room for overfit. The 2-Layer NN required ~18 epochs to converge, while the Deep NN benefited from its greater capacity and Dropout regularization, requiring approximately 25–30 epochs before validation loss plateaued.

Step	Operation	Tool/ Method	Justification
1	Data Loading	Pandas read_csv	Standard tabular ingestion
2	EDA	Matplotlib/ Seaborn	Understand distributions & imbalance
3	Null Check	df.isnull().sum()	Verify data integrity
4	Feature Scaling	StandardScaler	Normalize Time & Amount for NN
5	Stratified Split	train_test_split	Preserve fraud ratio in all splits
6	SMOTE	imbalanced-learn	Augment minority class (train only)
7	Class Weightscom	pute_class_weigh	t Penalize FN during training
8	Model Training	Keras / XGBoost	Fit models on augmented train set
9	Evaluation	sklearn metrics	AUC, Precision, Recall, F1 on test
10	Threshold Tuningr	oc_curve + Youden	Optimize operational threshold

Validation loss curves for the Deep NN showed initial rapid decrease followed by a plateau, with no signs of overfitting—confirming that Dropout and class weights effectively regularize the network despite the imbalanced training set. XGBoost converged in approximately 180 rounds (of 300 maximum), with monotonically increasing validation AUC before early stopping triggered at round 200.

E. Feature Importance (XGBoost)

XGBoost's gain-based feature importance analysis revealed the top discriminative features for fraud detection:

TABLE V. XGBoost Feature Importance (Gain-based)

V14 emerges as the single most discriminative feature, consistent with prior literature [3] on this dataset. The Time feature ranks near the bottom, confirming that absolute transaction time is less informative than the PCA-derived behavioral features, though temporal patterns visible in EDA may be better captured through engineered time-delta features in future work.

D. Complete Preprocessing Pipeline Summary

Table VI summarizes the complete preprocessing pipeline with key design decisions and their justification, providing a concise reference for practitioners implementing similar systems.

TABLE VI. Complete Preprocessing & Modeling Pipeline

Rank	Feature Im	portance (Gain	) Interpretation
1	V14	18.7%	Strongest fraud indicator
2	V4	14.2%	High positive correlation with fraud
3	V11	11.5%	Key discriminative PCA component
4	V17	9.8%	Moderate fraud association
5	Amount	6.3%	Transaction value signal
6–11	V01,V3,V7,V10,V12	~25%	Combined moderate importance
11–30	Remaining V, Time	~14.5%	Low individual importance

All experiments were conducted in Python 3.10 using the following library versions: TensorFlow 2.13, XGBoost 1.7.6, scikit-learn 1.3.0, imbalanced-learn 0.11.0, pandas 2.0, and NumPy 1.24. Experiments were run on a standard machine with Intel Core i7 processor and 16 GB RAM; no GPU acceleration was required for the models described in this paper given the dataset size.

Training time benchmarks: Sigmoid Baseline (~2s), 2-Layer NN (~45s), Deep NN (~3min), XGBoost (~4min). All random operations used random_state=42 for reproducibility. The complete pipeline from raw CSV to evaluated model required approximately 12 minutes end-to-end on the hardware described above.

A critical implementation note regarding EarlyStopping: the parameter restore_best_weights=True must be explicitly set in Keras; without it, the model weights at the final (potentially degraded) epoch are retained, not the optimal weights. This subtlety is frequently overlooked in tutorial implementations and can meaningfully degrade final model performance.

```
from tensorflow.keras.callbacks import
( EarlyStopping, ReduceLROnPlateau)
es = EarlyStopping(
monitor='val_loss',
patience=5,
restore_best_weights=True, # Critical!
verbose=0)
rlr =
ReduceLROnPlateau( monitor='val_
loss', factor=0.5, patience=3,
min_lr=1e-6)
history = model.fit(X_res, y_res,
validation_data=(X_val, y_val),
callbacks=[es, rlr],
class_weight=class_weights,
epochs=100, batch_size=2048,
```

6. DISCUSSION

A. Impact of SMOTE on Model Performance

Ablation experiments (models trained without SMOTE) showed that all four models experienced significant recall degradation for the fraud class. The Sigmoid Baseline without SMOTE achieved only 62% recall (vs 84.7% with SMOTE), while the Deep NN dropped to 71% recall (vs 93.9%). This confirms that SMOTE's synthetic sample generation is essential for enabling models to learn adequate decision boundaries for the minority class.

Critically, applying SMOTE after splitting and before test set construction (data leakage scenario) inflated reported recall values by 8–15 percentage points across all models—values that failed to materialize on genuinely unseen data. This underscores that leakage-free pipelines are not merely a methodological nicety but a practical necessity for trustworthy evaluation.

B. XGBoost vs. Deep Learning

XGBoost's superiority over the deep neural networks (AUC 0.991 vs 0.984) aligns with the broader observation in the machine learning community that gradient-boosted trees consistently outperform deep learning on tabular datasets, where feature interactions are sparse and the number of samples—while large in absolute terms—provides limited diversity in the minority class.

The Deep NN's competitive performance (AUC 0.984) demonstrates that with appropriate architecture design (funnel structure), regularization (Dropout), and imbalance handling (SMOTE + class weights), deep learning can be a viable alternative where end-to-end differentiable pipelines or feature learning capabilities are required.

C. Practical Deployment Considerations

In production deployment, several additional considerations arise. First, classification threshold tuning: the default 0.5 decision threshold may not be optimal; threshold selection based on business cost matrices (cost of FN vs FP) can further optimize the precision-recall tradeoff. Second, concept drift: fraud patterns evolve continuously, necessitating periodic model retraining or online learning approaches. Third, inference latency: XGBoost offers sub-millisecond inference on standard hardware, making it directly deployable in real-time transaction scoring pipelines without specialized infrastructure.

D. Ethical Considerations

Fraud detection models carry ethical implications. False positives disproportionately affect certain demographic groups if model features correlate with protected attributes. Since V1–V28 are PCA-transformed from anonymized data, direct demographic bias is difficult to assess but cannot be ruled out. Future work should apply fairness auditing frameworks such as AI Fairness 360 to evaluate disparate impact across customer segments.

E. Comparison with Related Work

Comparing our results with prior literature on the same dataset provides important context. Dal Pozzolo et al. [11] reported AUC values between 0.95 and 0.98 using undersampling strategies with calibrated classifiers, while our SMOTE-based approach achieves AUC of 0.991 with XGBoost—a meaningful improvement attributable to the combination of synthetic oversampling and scale_pos_weight.

Fiore et al. [5] reported recall improvements of 5–8% using GAN-based oversampling compared to SMOTE on banking datasets, suggesting that future integration of adversarial synthetic generation may further improve our Deep NN's recall. However, the substantially increased training complexity and sensitivity to GAN hyperparameters make SMOTE the more practical choice for most production deployments.

Our results are competitive with the state-of-the-art reported in the Kaggle community for this dataset (best public kernel AUC ~0.979 for neural networks, ~0.989 for XGBoost), with our leakage-free methodology providing a more reliable benchmark.

F. Threshold Optimization

All models were evaluated at the default 0.5 decision threshold. In operational settings, the optimal threshold is determined by the business cost matrix: the relative cost of a missed fraud (FN) versus the cost of blocking a legitimate transaction (FP). For a typical bank where fraud losses are 50–100x the operational cost of a false block, the optimal threshold computed via Youden's J statistic (maximizing sensitivity + specificity - 1) was approximately 0.35 for XGBoost, improving recall from 95.2% to 97.1% at the cost of reducing precision from 94.7% to 89.3%. This threshold-tuning capability is a significant operational advantage of probabilistic classifiers over rule-based systems.

```
from sklearn.metrics import roc_curve
fpr, tpr, thresholds =
roc_curve( y_te,
xgb.predict_proba(X_te)[:,1])
```

```
j_scores = tpr - fpr
opt_thresh = thresholds[j_scores.argmax()]
print(f'Optimal threshold: {opt_thresh:.3f}')
y_pred_opt = (xgb.predict_proba(X_te)[:,1]
>= opt_thresh).astype(int)
```

G. Limitations

This study has several limitations: (1) The dataset's PCA transformation prevents interpretable feature analysis in the original feature space, limiting business insight extraction. (2) The 2013 dataset may not reflect contemporary fraud patterns such as card-not-present fraud, account takeover, or synthetic identity fraud that have grown significantly post-2013. (3) Our evaluation is limited to offline batch settings; real-time streaming evaluation under concept drift was not conducted. (4) Hyperparameter optimization was limited to manual search; systematic Bayesian optimization may yield further performance gains. (5) The computational cost of SMOTE scales with training set size, which may be prohibitive for very large financial datasets.

7. CONCLUSION AND FUTURE WORK

This paper presented a comprehensive, end-to-end credit card fraud detection pipeline addressing the fundamental challenge of extreme class imbalance through SMOTE oversampling, class weighting, and careful experimental design that prevents data leakage. Four models—Sigmoid Baseline, 2-Layer Neural Network, Deep Neural Network, and XGBoost—were rigorously evaluated on a held-out test set preserving real-world class distribution.

Key findings: (1) SMOTE applied exclusively to the training set is essential for achieving acceptable fraud recall, reducing the False Negative Rate by 3–10x compared to no oversampling. (2) XGBoost achieves the best overall performance (AUC=0.991, Recall=0.952, FN Rate=4.04%), outperforming all neural network architectures on this tabular dataset. (3) The Deep Neural Network (AUC=0.984) demonstrates that carefully designed deep learning architectures with Dropout regularization provide competitive alternatives where end-to-end differentiable pipelines are required. (4) Data leakage from premature SMOTE application inflates reported metrics by 8–15%, making leakage-free pipelines mandatory for trustworthy evaluation. (5) XGBoost's feature importance analysis confirms V14 as the most discriminative fraud indicator, consistent with domain knowledge from prior literature.

A. Summary of Contributions

The primary contributions of this work are: a production-grade, leakage-free preprocessing pipeline for imbalanced fraud detection; a systematic ablation study comparing four model families under identical conditions; detailed confusion matrix analysis with operational FN Rate metrics; and threshold optimization analysis connecting model output to business cost decisions. Together, these provide a complete, reproducible reference implementation for practitioners building fraud detection systems.

B. Future Directions

Future research will pursue multiple directions to extend this work:

- Graph Neural Networks (GNNs): Modeling cardholder-merchant-device transaction networks as heterogeneous graphs, capturing relational fraud patterns invisible to individual transaction features.
- Real-time Streaming Detection: Integration with Apache Kafka and Flink for sub-100ms fraud scoring in live payment processing pipelines, with online learning to handle concept drift.
- Federated Learning: Enabling multiple financial institutions to collaboratively train fraud models without sharing raw transaction data, preserving regulatory compliance and customer privacy.
- Transformer-based Sequence Models: Treating a cardholder's transaction history as a temporal sequence and applying attention mechanisms to detect behavioral anomalies over time.
- Fairness Auditing: Applying AI Fairness 360 to evaluate disparate impact of fraud predictions across demographic groups, ensuring equitable treatment of all customer segments.
- Automated Hyperparameter Optimization: Systematic Bayesian search over neural architecture hyperparameters (depth, width, dropout rates, learning rates) to identify globally optimal configurations.

The complete codebase using publicly available libraries (TensorFlow [10], XGBoost [4], imbalanced-learn [9], scikit-learn [8]) on the open Kaggle dataset [7] makes all results fully reproducible, providing a practical reference for practitioners building production-grade fraud detection systems in real-world financial environments.

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